DATA SCIENCE SUMMER TRAINING Batch-3

MINI PROJECT-2 (Analysis of IPL DATASET)

Project statement

Part A – Data Wrangling & Exploration (Pandas & NumPy)

All code must be accompanied by markdown cells explaining the reasoning.

1. **Load & Merge**
    - Load matches.csv and deliveries.csv into Pandas DataFrames.
    - Merge the ball-by-ball data with match data on match_id to create a single analytical DataFrame.
    - Display the shape, column names, and data types. How many seasons are covered?
2. **Handle Missing Data**
    - Identify columns with missing values.
    - For city (in matches), decide whether to drop or impute; justify your choice.
    - For player_of_match, how would you treat missing values without losing important match information? Implement it.
    - In the deliveries data, replace any NaN values in dismissal_kind with 'not out'.
3. **Feature Engineering (NumPy & Pandas)**
    - Create a new column total_runs which is the sum of batsman_runs and extra_runs.
    - Using NumPy, classify a delivery as a **boundary** if batsman_runs >= 4, else 'non-boundary'. Store in a new column is_boundary.
    - Extract the **over number** from the over column (e.g., 1.1 → 1st over). Use this to create a categorical column over_phase:
        - Powerplay: overs 1-6
        - Middle: 7-15
        - Death: 16-20
    - From the date column, derive year and month for seasonal trend analysis.
4. **Aggregations & Summary Statistics**

- Compute and display the top 5 batsmen by total runs scored across all seasons.
- Find the economy rate (runs conceded per over) of each bowler who has bowled at least 500 balls. Use NumPy to calculate the economy rate efficiently.
- For each season, determine the team with the highest win percentage. Show the results as a DataFrame.

Part B – Visual Analysis & Pattern Discovery (Matplotlib & Seaborn)

Every plot must have a clear title, labeled axes, and a legend where appropriate. After each plot, add a markdown cell with 2-3 sentences interpreting what the chart reveals.

5. **Toss Impact on Match Outcome**
    - Create a grouped bar chart (Seaborn) showing the percentage of matches won when choosing to bat first vs. field first, across all seasons. Does the toss decision show a consistent advantage? Interpret.
6. **Top Run-Scorers & Their Dismissal Patterns**
    - Plot a horizontal bar chart of the top 10 run-scorers.
    - For the top 3 run-scorers, create a stacked bar chart (Seaborn) showing the count of different dismissal kinds (caught, bowled, lbw, etc.). What does this say about their vulnerabilities?
7. **Seasonal Batting Trends**
    - Using a line plot, show the average runs per match (total runs / total matches) for each season. Highlight any noticeable spikes or dips.
    - Overlay a trend line (regression) using Seaborn's lmplot or regplot on a scatter plot of season vs. average runs per match. Is the IPL becoming more high-scoring over time?
8. **Match Outcome & Venue Analysis**
    - Select the top 5 stadiums by number of matches hosted. For these stadiums, create a count plot of the match winner (or a heatmap of win counts by team). Which venue sees the most "home advantage"?
9. **Death Over Performance**
    - Filter the deliveries DataFrame for overs 16–20.
    - Plot a box plot (Seaborn) of runs scored per over during death overs, grouped by batting team (choose top 6 teams by total death-over runs).

- Which team scores most consistently in death overs, and which is most explosive (high median & high variance)? Interpret.

Part C – Storytelling & Final Report

Combine your analyses into a 2–3 page report (PDF or Jupyter Notebook) structured as follows:

1. **Executive Summary** – What is the most important insight from your analysis?
2. **Key Insights** – Present 4–5 bullet-pointed findings, each supported by a visualization and a brief interpretation. For example:
   "Chennai Super Kings have the highest win percentage while chasing after winning the toss, suggesting a strategic preference for known targets."
3. **Recommendation for Team Strategy** – Based on your findings, what one concrete piece of advice would you give to a team captain/coach regarding toss, batting order, or venue selection?
4. **Self-Reflection** – Briefly describe one limitation of the dataset that prevented deeper analysis, and what additional data you would have liked.